

RetailGPT: Technical Overview

Enterprise Decision Intelligence Platform

1. Project Summary

RetailGPT is an AI-powered retail analytics platform featuring an Azure-emulated data pipeline, LangChain orchestration, and a premium FastAPI/Chart.js dashboard.

2. Core Architecture

Backend: FastAPI with modular routing and SQLite database.

Frontend: Native HTML/CSS/JS with Chart.js 4.4 visualization.

Data Layer: Parquet-based curated storage with Azure-style engineering pipeline.

3. Key Features

Feature	Description
Dynamic Dashboard	Interactive charts with auto-handling for empty states.
AI Copilot	Multi-agent system (ML, RAG, Data) via LangChain.
MBA Engine	Market Basket Analysis using Apriori algorithm.
Forecast & Anomalies	LightGBM forecasting with Isolation Forest detection.

4. Recent Enhancements

- **Automated KPIs:** Dashboard now fetches real-time sales and growth metrics.
- **Intelligent Empty States:** Full-page guidance when no data is detected.
- **Visualization Overhaul:** Clean line-based sales charts with anomaly markers.
- **Data Robustness:** Enhanced null safety and dynamic scaling for analytics.